

# Shear profiles inside gas sparged submerged hollow fiber membrane modules

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## Abstract

Gas sparging is a common strategy used to minimize and prevent fouling on membrane surfaces. Although there have been numerous studies on the hydrodynamic conditions and the mechanisms of fouling control in confined membrane systems such as tubular membranes, there is very limited literature available that describes the hydrodynamic conditions and mechanisms of fouling control in unconfined systems such as submerged hollow fiber membranes. The present study investigated bubble-induced shear profiles in submerged hollow fiber membrane modules using an electrochemical method for measuring shear forces. The effects of different operating conditions such as gas sparging rates, fiber packing densities, diffuser nozzle sizes, and module configurations (i.e. loosely versus tightly held) on shear profiles were studied. It was found that the hydrodynamic conditions in confined tubular membrane systems and unconfined submerged hollow fiber membranes are different, which implies that the mechanisms of fouling control in submerged hollow fiber membrane units may be different than those in confined systems such as the tubular membranes.

Surface shear profiles during sparging were found to be substantially affected by the operating conditions (i.e. gas sparging rate, fiber packing density, diffuser nozzle size, and module configuration). The different shear profiles were due to the different bubble geometries and flow paths which resulted from the different operating conditions considered. For tightly held low and medium fiber packing density bundles the high number of shear events generated by small bubbles (i.e. small diffuser nozzle) resulted in more favorable hydrodynamic condition for fouling reduction and flux enhancement. The higher permeate fluxes that have been reported in loosely held submerged hollow fiber membrane systems likely result from the larger number of fibers (i.e. regions) that can be impacted by sparged bubbles in systems with loosely held fibers.

The results from the present study provide a better understanding of the hydrodynamic conditions inside submerged hollow fiber membrane modules, and therefore a better understanding of possible mechanisms of fouling control via gas sparging.

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**Keywords:** Submerged hollow fiber membrane; Fouling control; Gas sparging; Shear profiles; Electrochemical method

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## 1. Introduction

Over the past decade, the use of microfiltration and ultra-filtration processes has been increasing in popularity for water and wastewater treatment applications, primarily due to increasing public health concerns with respect to potable water quality, as well as the emergence of lower cost membranes. Moreover, compared to conventional technologies, membrane filtration processes offer the advantages of reducing the amount of treatment chemicals needed, requiring a smaller footprint, and reducing labor requirements since the processes can be

easily automated. Despite the emergence of new and lower cost membranes, the unavoidable problem of membrane fouling remains associated with membrane filtration processes. Membrane fouling causes a reduction in total permeate yield, increases operational difficulties and ultimately increases the overall operational and capital costs associated with membrane processes.

Many strategies have been developed to control and minimize membrane fouling. One such strategy is the control of hydrodynamic conditions near membrane surfaces to prevent particle deposition on the membranes, thereby reducing the degree of membrane fouling and enhancing permeate flux. Gas sparging at the bottom of the membrane units is commonly used to achieve these favorable hydrodynamic conditions. Gas sparging is known to enhance permeate flux by back-transport of

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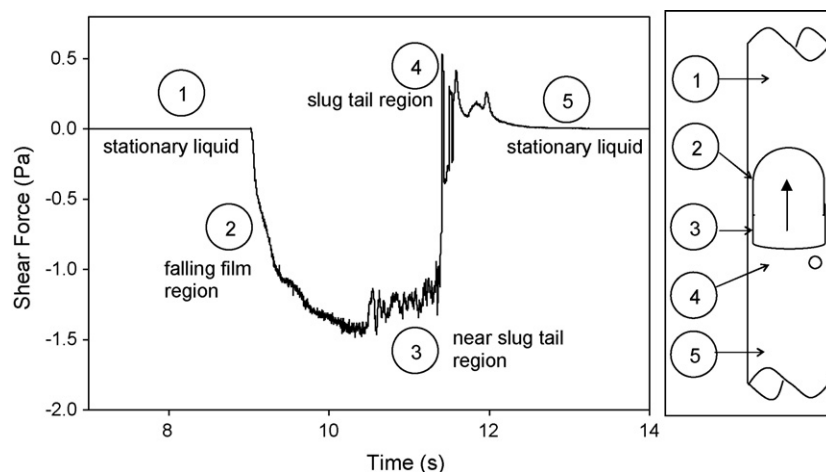


Fig. 1. Wall shear stress distribution around a gas slug rising in stationary liquid in a vertical pipe, as measured experimentally using the electrochemical shear method [11].

material (i.e. foulant) away from membrane surfaces [1–6]. The mechanisms by which gas sparging increases the back-transport of material away from membrane surfaces may depend on the membrane configuration used (i.e. confined versus unconfined membranes) [7]. Confined modules, such as tubular membranes, are modules in which the feed water and sparged gas bubbles are confined by the membranes in an inside-out flow configuration. Unconfined modules, such as submerged hollow fibers or flat sheet membranes, exhibit an outside-in flow configuration. The degree of fouling reduction in a confined membrane module is dependent on the bubble dynamics inside the confined membrane. Slug flow has been found to minimize fouling on membrane surfaces to a greater extent than bubble, churn and annular mist flows in confined membrane systems [2,4,8–10]. The mechanism of fouling reduction and flux enhancement for slug flow inside a confined membrane system has been linked to the high shear forces induced by liquid flow reversal in the falling film surrounding a rising gas slug, the oscillating flows near the tail end of the rising gas slug, and the overall increase in the bulk superficial velocity imparted by the rising gas slug [4]. A typical shear profile of slug flow in stationary liquid inside a confined vertical tube is presented in Fig. 1 [11].

As for gas sparged unconfined systems such as submerged hollow fiber membranes, very limited experimental literature exists on the mechanisms by which fouling reduction and flux enhancement is achieved. When studying the effect of gas sparging on submerged hollow fibers, some researchers have assumed that since the packing density of the fibers is high, the surface shear forces induced by gas bubbles rising between fibers are similar to those induced by gas slugs in a confined system, and as a result, the mechanism of flux enhancement in unconfined membrane systems has often been assumed to be similar to that of confined membrane systems [4,12]. The validity of this assumption is questionable, given the complex and transient nature of the geometry and the hydrodynamic conditions in submerged membrane modules. Depending on the aeration system, bubbles with different sizes and shapes can form in an unconfined system (i.e. spherical bubble, ellipsoidal bubble or spherical cap bub-

bles), and each bubble geometry yields different rise velocities and subsequently results in different shear forces acting on the membrane surface. Loosely held fibers in the submerged hollow fiber system may sway laterally as a result of the wakes generated by the rising bubbles. The hydrodynamic conditions surrounding gas sparged submerged hollow fibers are poorly understood. This knowledge gap presents a real impediment to developing a complete model of the filtration processes and to optimizing the performance of submerged hollow fiber membranes in terms of membrane geometry and energy costs associated with bubbling.

The objective of the present study is to address the knowledge gap discussed above from the point of view of hydrodynamics. More specifically, the question of whether the hydrodynamic conditions near the membrane surfaces of unconfined systems are similar to those of confined systems was addressed. The directional surface shear forces generated by rising bubbles in tightly and loosely held fiber bundles was measured for different fiber packing densities, diffuser nozzle sizes and gas sparging rates.

## 2. Materials and methods

Surface shear forces generated by rising bubbles can be measured electrochemically [13,14]. Recently, with the realization of the importance of shear forces near membrane surfaces in controlling fouling, a small number of researchers have used electrochemical probes to measure the wall shear stresses on membrane surfaces during filtration [16].

However, these studies focused on confined membrane systems. For unconfined systems, Ducom et al. [15] used the electrochemical shear probe to study shear forces acting on flat sheet membranes under gas sparging, and Bérubé et al. [17] studied the shear forces acting on submerged hollow fiber membranes under gas sparging conditions. However, none of the electrochemical probes used in their study could provide information regarding the direction (i.e. sign) of the surface shear forces. An electrochemical probe, similar to those developed by Cognet et al. [18] was developed in the present study to measure

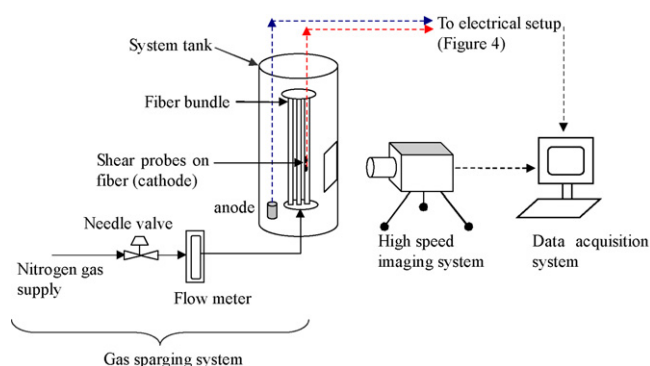


Fig. 2. The schematic diagram of the experimental set-up.

both the direction and magnitude of the surface shear force on gas sparged submerged hollow fibers.

The experimental system consisted of a shear probe, an electrolytic solution, a data acquisition system, fiber modules, a system tank, a gas sparging system, and a high speed imaging system. The schematic diagram of this set-up is shown in Fig. 2. The experimental program was based on a factorial-type design which considered different module configurations (loosely and tightly held), fiber packing densities (8, 16 and 31 fibers/cm<sup>2</sup>), diffuser nozzle sizes (1 and 2 mm) and gas sparging rates (2, 10 and 35 mL/min).

### 2.1. Double-probe for shear force measurement

A 1 m long Teflon tube with similar dimension and flexibility as a hollow fiber membrane (o.d. 1.6 mm) was used as the “test fiber”. The shear probe, which consisted of two cathodes placed side by side, were embedded in the test fiber to measure shear forces acting on the fiber surface under different flow conditions. A magnified view of the shear probe embedded in the test fiber is shown in Fig. 3. Each cathode was made from a small piece of platinum wire, and the surface of the cathodes were circular with diameters of 0.5 mm. The platinum wire was coated with epoxy. As a result, when placed side by side, the spacing between the cathodes was approximately 0.03 mm. A 1 mm hole was drilled onto the surface of the test fiber through which the shear probe was inserted from the underside of the test fiber (Fig. 3b). The hole with the protruding shear probe was filled with epoxy, and the shear probe was sanded flush with the surface of the test fiber

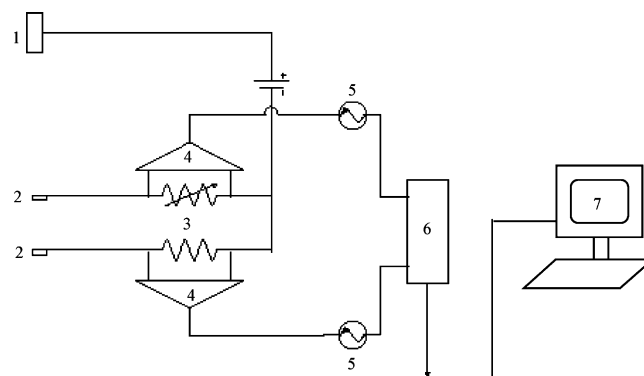


Fig. 4. The electrical set-up of the experiment. (1) Anode, (2) cathodes/shear probe, (3) 100 ohm resistor, (4) 1K amplifier, (5) low pass filter, (6) data acquisition box and (7) computer with Labview.

using a jeweler’s file. After fabrication, each of the test fibers with a shear probe were examined under a stereomicroscope to ensure that the two cathodes were not in contact with each other and the probe spacing was less than 0.05 mm. Two cathodes are required to measure the direction as well as the magnitude of the shear forces as discussed in the following section.

### 2.2. Electrolytic solution and data acquisition system

The electrolyte solution used for the experiments contained 0.003 M of ferricyanide, 0.006 M ferrocyanide and 0.3 M potassium chloride in de-ionized water. De-ionized water was purged with nitrogen gas to de-oxygenate the water, since oxygen can oxidize the ferricyanide and ferrocyanide compounds. The diagram of the electrical set-up is shown in Fig. 4. The cathode of the electrochemical system was the shear probe, and the anode was a cylindrical piece of nickel (3 cm long  $\times$  2 cm diameter). Each cathode in the shear probe had its own independent electrical circuit. A 250 mV power supply provided the driving force for the electrochemical reaction between the anode and the cathode. A shear probe signal was obtained via voltage drop across the 100 ohm resistor.

This signal (in V) was then conditioned through an amplifier and a low pass filter. A data acquisition box with two different channels acquired the signals from the cathodes, and real time data was recorded at a rate of 1000 Hz using Labview. If the signal of the upstream cathode was higher than that of the

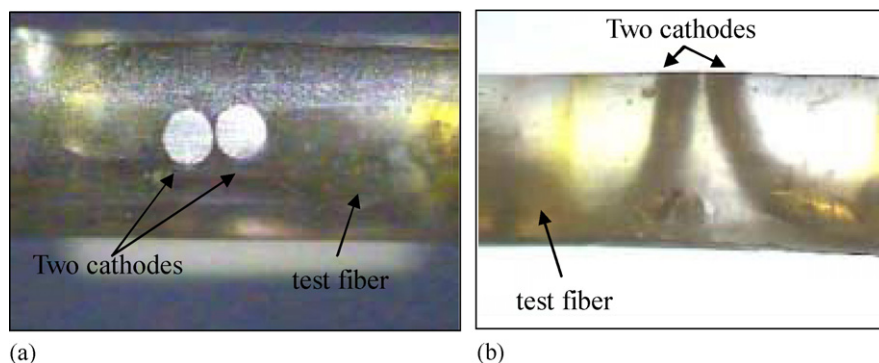


Fig. 3. Test fiber with embedded shear probe. (a) Top view and (b) side view.

downstream cathode, the signal was deemed to be positive, and the shear signal value from the upstream cathode was taken as the shear signal of the system [18]. On the other hand, if the shear signal of the downstream cathode was higher than the signal of the upstream cathode, then the sign of the shear signal was assumed to be negative (representing flow reversal) and the value of the shear signal was that measured from the downstream cathode.

For each experiment, data was collected for a period of 10 s, and data was acquired at a rate of 1000 Hz, generating a total of 10,000 data points. Each data point was referred to as a “frame number”, which corresponds to the image number captured using a high-speed video camera. In the present study, the shear force measurements are presented in units of volts, instead of force per unit area (i.e. Pa). To convert voltage measurements to shear forces, it must be assumed that (1) the axial diffusion is negligible, (2) the velocity gradient in the direction normal to the electrode surface is negligible, (3) a linear velocity gradient in the region of measurement (in the thin concentration boundary layer) exists [19], and (4) the cathode is circular. Although these assumptions are likely valid for the experimental set-up used in the present study, voltage signals are presented to avoid introducing any bias into the experimental results. Equations used to convert voltage signals to shear forces are found elsewhere [17] for the experimental set-up used in the present study.

### 2.3. Fiber modules and system tank

For all experiments, the hollow fibers used consisted of Teflon tubes with similar dimensions and flexibility as that of commercially available hollow fiber membranes (o.d. 1.6 mm). Teflon tubes were used instead of actual hollow fiber membranes because the Teflon tubes are translucent, which enabled images of rising bubbles within the submerged fibers to be captured by a high-speed camera. The Teflon tubes are referred to as “fibers” from here onwards. The length of each fiber in a bundle was 12 cm, and the fibers were held in place at both ends by a circular disk, as presented in Fig. 5. Three fiber-packing densities

were studied: low, medium, and high fiber packing densities (8, 16 and 32 fibers/cm<sup>2</sup>, respectively). All fibers were arranged in a square matrix. For the bundle with low fiber packing density, four fibers, including the test fiber, were inserted into the holes in the circular disks to form a square matrix. Each fiber was separated by an empty 1.6 mm hole in the circular disk, as shown in Fig. 5a. The resulting distance between each fiber was 3.6 mm. For the bundle with medium fiber packing density, eight fibers, including the test fiber, were inserted into the holes in the circular disks (Fig. 5b). The distance between fibers was 1 mm. For the bundle with high fiber packing density, eight fibers were placed side-by-side in a square matrix with no space between each fiber (except for a hole in the middle of the bundle for the diffuser nozzle), and glued together by epoxy at both ends (Fig. 5c). A circular disk with a large hole drilled in the middle was used to hold this bundle at both ends. In all of the arrangements, the shear probe was positioned at mid length of the fiber (i.e. 6 cm from the bottom of the bundle). The shear probe on the test fiber was oriented such that the surface of the probe faced the diffuser nozzle (i.e. center of the fiber bundle).

The fiber bundles were immersed in a cylindrical system tank filled with the electrolyte solution. Both ends of the bundle (i.e. circular discs) were held in place such that the bundles could be held either loosely or tightly by adjusting the height at the top of the bundle. For tightly held bundles, the distance between the circular disks which held the ends of the fibers was 12 cm. For loosely held fibers, the distance between the circular disks was 5% (i.e. 11.4 cm).

The cylindrical system tank was constructed of plexiglas so that bubbles in the tank could be observed visually. The system tank was sealed at the top to prevent oxygen exposure during each experiment. The nickel anode was also immersed in the system tank.

### 2.4. Gas sparging system

Nitrogen gas was delivered to the center of the bundles using diffuser nozzle sizes of 1 or 2 mm (i.d.). Nitrogen gas, instead

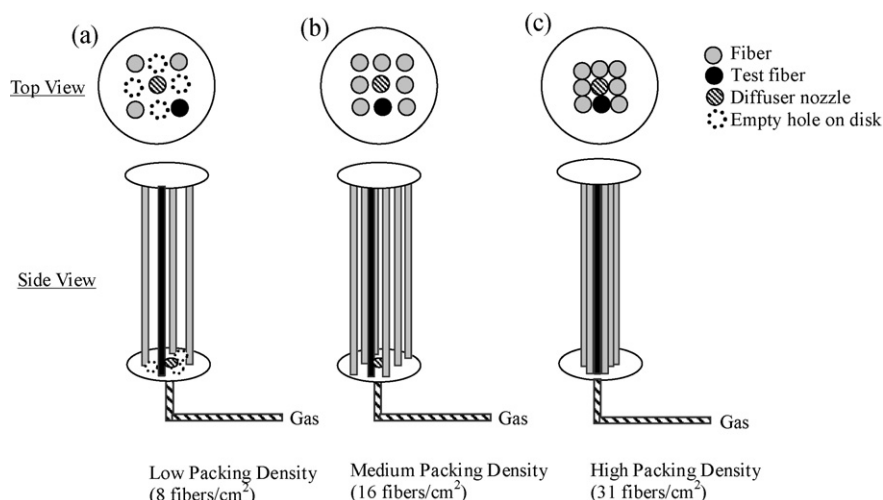


Fig. 5. Top and side views of the fiber arrangements in the bundles with different fiber packing densities.

of air, was used because ferricyanide and ferrocyanide reacts chemically with oxygen. Three different gas sparging rates were investigated: 2, 10 and 35 mL/min.

### 2.5. High speed imaging system

A high-speed video camera (Kodak Ektapro 1000 Motion Analyzer System) was used to take images of the sparged bubbles as they rose through the fiber bundles. The high-speed camera captured images at a rate of 1000 Hz, which was the same frequency at which the shear signals were acquired. The camera was connected to the data acquisition system, so that the acquisition of both the shear signals and the video camera images were synchronized. The error in synchronization was determined to be less than 1 Hz. A small section of the cylindrical system tank was cut out and replaced by a flat panel made of plexiglas so that images taken by the video camera were not optically distorted by the curvature of the cylindrical tank.

### 2.6. Experimental program

An experimental program with a factorial design which considered different module configurations (loosely and tightly held), fiber packing densities (8, 16 and 31 fibers/cm<sup>2</sup>), diffuser nozzle sizes (1 and 2 mm) and gas sparging rates (2, 10 and 35 mL/min) was conducted to quantify shear forces acting on gas sparged submerged hollow fiber modules. Each experiment was repeated three times.

In a separate series of experiments, the test fiber was rotated clockwise and counter clockwise such that the shear probes were facing away from the rising bubbles (i.e. center of the fiber bundle). These experiments were performed with a fiber packing density of 16 fibers/cm<sup>2</sup>. A zero degree rotation indicated that the shear probe was facing the rising bubble, while a 180° rotation indicates that the probes were facing away from the rising bubble.

All experiments were conducted at room temperature (25 °C) and under conditions of limiting current (results not presented). A summary of all experiments performed is provided in Table 1. The first, number in the notation used to name the different experiments refers to diffuser nozzle size used (i.e. 1 or 2 mm). The subsequent letter and number correspond to the fiber packing density (i.e. A, B or C for low, medium or high fiber packing densities, respectively) and the gas sparging rate (i.e. 1, 2 or 3 for 2, 10 or 35 mL/min, respectively), respectively. The Rotate notation corresponds to the experiments where the test fiber was rotated clockwise and counter clockwise, and the 1 and 2 notation correspond to gas sparging rates of 5 and 30 mL/min, respectively.

## 3. Results

### 3.1. Bubble-induced shear profile for tightly held fiber bundles

Typical shear signals obtained for the tightly held, medium packing density fiber bundle at different gas sparging rates (2,

Table 1  
Summary of the experimental program

| Experiment number | Fiber packing density | Diffuser nozzle size (mm) | Gas sparging (mL/min) | Fiber configuration                |
|-------------------|-----------------------|---------------------------|-----------------------|------------------------------------|
| 1-A1              | Low                   | 1                         | 2                     | Tightly held                       |
| 1-B1              | Med                   | 1                         | 2                     | Tightly held                       |
| 1-C1              | High                  | 1                         | 2                     | Tightly held                       |
| 2-A1              | Low                   | 2                         | 2                     | Tightly held                       |
| 2-B1              | Med                   | 2                         | 2                     | Tightly held                       |
| 2-C1              | High                  | 2                         | 2                     | Tightly held                       |
| 1-A2              | Low                   | 1                         | 10                    | Tightly held                       |
| 1-B2              | Med                   | 1                         | 10                    | Tightly held                       |
| 1-C2              | High                  | 1                         | 10                    | Tightly held                       |
| 2-A2              | Low                   | 2                         | 10                    | Tightly held                       |
| 2-B2              | Med                   | 2                         | 10                    | Tightly held                       |
| 2-C2              | High                  | 2                         | 10                    | Tightly held                       |
| 1-A3              | Low                   | 1                         | 35                    | Tightly held                       |
| 1-B3              | Med                   | 1                         | 35                    | Tightly held                       |
| 1-C3              | High                  | 1                         | 35                    | Tightly held                       |
| 2-A3              | Low                   | 2                         | 35                    | Tightly held                       |
| 2-B3              | Med                   | 2                         | 35                    | Tightly held                       |
| 2-C3              | High                  | 2                         | 35                    | Tightly held                       |
| 1-A3              | Low                   | 1                         | 35                    | Loosely held                       |
| 1-B3              | Med                   | 1                         | 35                    | Loosely held                       |
| 1-C3              | High                  | 1                         | 35                    | Loosely held                       |
| 2-A3              | Low                   | 2                         | 35                    | Loosely held                       |
| 2-B3              | Med                   | 2                         | 35                    | Loosely held                       |
| 2-C3              | High                  | 2                         | 35                    | Loosely held                       |
| Rotate 1          | Med                   | 2                         | 5                     | Tightly held, rotate –180° to 180° |
| Rotate 2          | Med                   | 2                         | 30                    | Tightly held, rotate –180° to 180° |



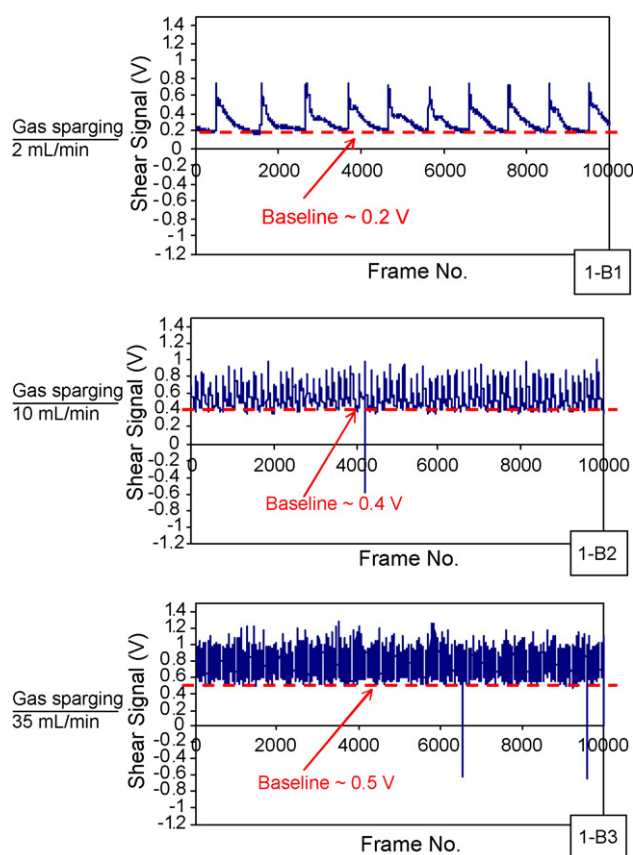


Fig. 6. Typical shear signals on test fiber at a medium fiber packing density and different gas sparging rates for tightly held fibers. Diffuser nozzle size: 1 mm.

10 and 35 mL/min) and diffuser nozzle sizes of 1 and 2 mm are shown in Figs. 6 and 7. The shear profiles for the low and high packing density experiments are shown in Figs. A.1 and A.2 in Appendix A.

As expected, for all experiments, increasing the gas sparging rate increased the “baseline” shear signals. The baseline shear signal obtained for the experiment with a medium fiber packing density and a gas sparging rate of 2 mL/min (graph 1-B1, Fig. 6) was approximately 0.2 V. For the same bundle, increasing the gas sparging rate to 10 and 35 mL/min increased the baseline shear signals to 0.4 and 0.5 V, respectively (graphs 1-B2 and 1-B3, Fig. 6). The increase in the baseline shear signals resulted from an increase in the bulk superficial liquid velocity at the higher gas sparging rates. However, the contribution of the bulk superficial liquid velocity to overall surface shear force is likely quite small when compared to that of the instabilities induced by rising bubbles. As presented in Figs. 6 and 7 (and Figs. A.1 and A.2 in Appendix A), the sparged bubbles induced instabilities which generated relatively large peaks in the shear signals. The peak shear signals induced by the instabilities were typically two to five times higher than those associated with the baseline shear signals. A detailed discussion on the causes of the peaks in shear signal, and their significance to flux enhancement is presented in the following sections.

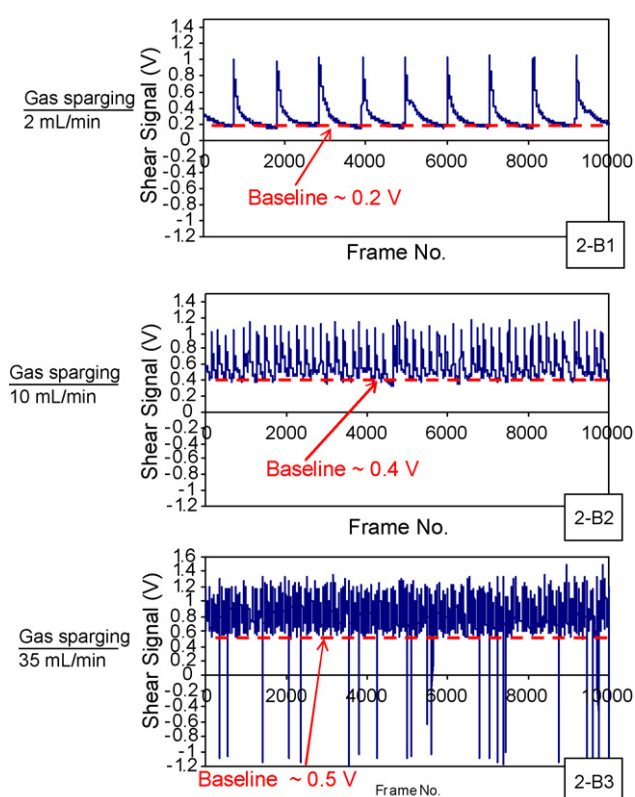


Fig. 7. Typical shear signals on test fiber at a medium fiber packing density and different gas sparging rates for tightly held fibers. Diffuser nozzle size: 2 mm.

### 3.1.1. Number of shear events, and magnitude of shear signals

In Figs. 6 and 7, each peak in the shear signal was considered to be a shear event caused by the passage of a rising gas bubble. The number of shear events observed for different fiber packing densities, diffuser nozzle sizes and gas sparging rates is summarized in Fig. 8. For all experiments, increasing the gas sparging rate increased the number of shear events. This was expected since the number of gas bubbles introduced into the system increases as the gas sparging rate increases.

When comparing the number of shear events obtained for the three different fiber packing densities, the experiments performed with medium fiber packing density generally resulted in the highest number of shear events, compared to those with low and high fiber packing densities. For low fiber packing densities, the rising bubbles were not completely and consistently confined by the fibers, especially at higher gas sparging rates (i.e. 10 and 35 mL/min). As a result, some bubbles periodically escaped from the bundles through the relatively large gaps that existed between the fibers. The non-confining nature of the low fiber packing density bundle lowered the number of shear events. This phenomenon can be seen in graph 2-A2 of Figs. A.1 in (in Appendix A), where no shear events were observed between frames 5000 and 7000 as a result of bubbles escaping the bundle before reaching the shear probe. A different phenomenon occurred to generate a relatively low number of shear events for bundles with high fiber packing density. Due to the high fiber packing density, the bubble break-off rate at the diffuser nozzle

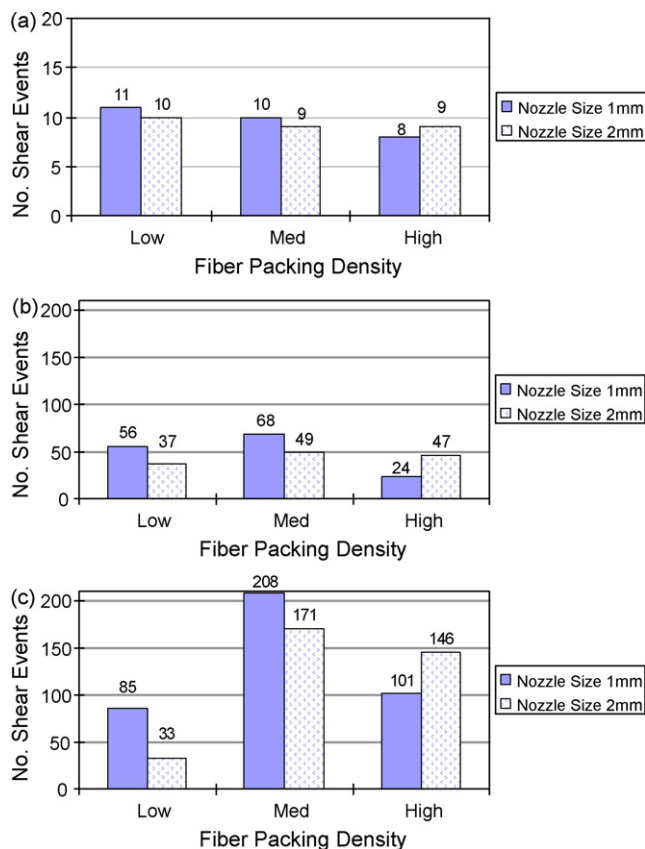


Fig. 8. The number of shear events (over a period of 10 s) experienced by tightly held fibers during gas sparging using diffuser nozzle sizes of 1 and 2 mm. (a) Gas sparging rate 2 mL/min, (b) gas sparging rate 10 mL/min and (c) gas sparging rate 35 mL/min. Note the scale for (a) is different than that of (b) and (c).

was less than that for the bundles with low and medium fiber packing densities. As a result, fewer and larger bubbles, resembling gas slugs rising in a confined system, were observed for high fiber packing density bundles. Typical images of bubbles rising in medium and high fiber packing density bundles are shown in Fig. 9a and b, respectively.

As presented in Fig. 8, for experiments with the low and medium fiber packing densities, the number of shear events obtained was consistently higher when sparging with a 1 mm diffuser nozzle when compared to sparging with the 2 mm dif-

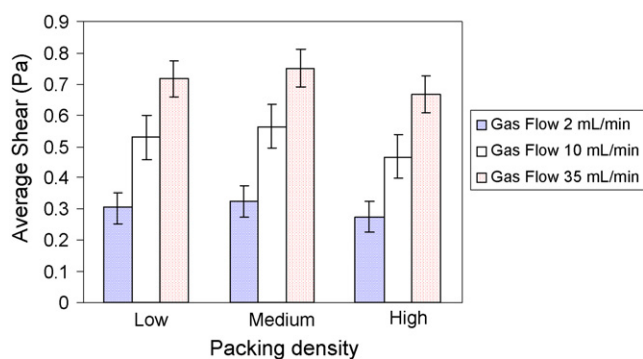


Fig. 10. Average shear signal for each experiment with diffuser nozzle size of 1 mm.

fuser nozzle. For experiments with high fiber packing density, however, the opposite trend was observed. The reason for this discrepancy was that bubbles delivered through the 2 mm diffuser nozzle broke apart to form smaller bubbles when they came into contact with the surrounding closely packed fibers. On the other hand, frequent bubble break-up was not observed for bubbles delivered through the 1 mm diffuser nozzles. As a result, fewer and larger bubbles were observed for high fiber packing density bundles. Typical images of the bubbles delivered through the 1 and 2 mm diffuser nozzle for high fiber packing density bundles are shown in Fig. 9b and c, respectively. The average shear signals for experiments performed with a diffuser nozzle of 1 mm and different gas sparging rates and fiber packing densities are presented in Fig. 10. As shown, increasing the gas sparging rate increased the average shear signals. However, increasing the fiber packing density did not have a significant impact on the average shear signal. Also, sparging with a larger diffuser nozzle (i.e. 2 mm) did not have a significant impact on the average shear signal (results not shown).

A number of studies have reported that gas sparging with small bubble diffusers (i.e. small bubbles) is better at controlling fouling in submerged hollow fiber membrane systems, when compared to sparging with coarse bubble diffusers [20,21]. Considering that the size of the diffuser nozzle used did not significantly impact the magnitude of the average shear signal, the results from the present study suggest that the better fouling control associated with sparging with smaller bubbles is likely

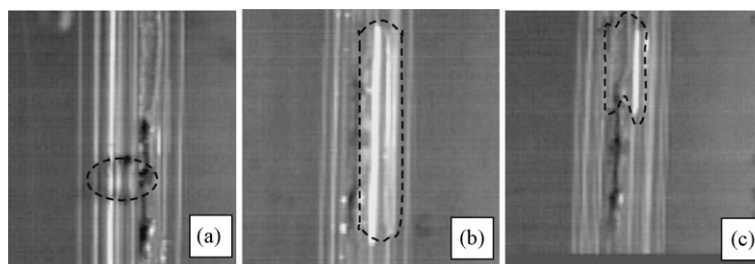


Fig. 9. Typical images of bubbles rising in fiber bundles. The translucent long streaks seen in the pictures are the eight fibers in the bundle. The black dashed line highlights the bubbles rising within the fiber bundle. (a) Medium fiber packing density, gas sparging 35 mL/min, diffuser nozzle size 1 mm, tightly held fibers. The bubble was approximately spherical. (b) High fiber packing density, gas sparging 35 mL/min, diffuser nozzle size 1 mm, tightly held fibers. The bubble delivered through the 1 mm diffuser nozzle is longer than the bubble in 'c'. (c) High fiber packing density, gas sparging 35 mL/min, diffuser nozzle size 2 mm, tightly held fibers. The bubble is longer than the bubble in 'b'.

related to the number of shear events that the fibers experience. The higher number of shear events generated by small bubbles (i.e. small diffuser nozzle) in low and medium fiber packing density bundles likely provides more favorable hydrodynamic condition for fouling reduction and flux enhancement.

### 3.1.2. Shear profile during shear event

As presented in Fig. 9, the geometry of the bubbles rising between fibers was substantially affected by the experimental conditions (i.e. diffuser nozzle size, gas sparging rate and fiber packing density). The effect of bubble geometry and flow path on the resulting shear signals is discussed in the sections below.

#### 3.1.2.1. Contribution of rising bubbles to positive peak shear forces.

For the experiments in which the shear signals were consistently positive (e.g. graph 1-B1, Fig. 6), the positive shear peaks were caused by the wake generated at the tail end of a bubble as the bubble rose past the shear probe. Fig. 11 shows a typical shear profile of a tightly held fiber bundle during the passage of a bubble, and the corresponding bubble images. As the bubble approached the shear probe, the shear signal began to rise rapidly (points 1 and 2, Fig. 11). When the tail end of the bubble passed by the shear probe, a positive peak in the shear signal occurred (point 3, Fig. 11). The decrease in shear signal after this

point occurred relatively slowly, even though the bubble was no longer in the vicinity of the shear probe (point 4, Fig. 11). These results suggest that the wake region of the rising bubble was long, and that the secondary flows in the wake region likely play an important role in contributing to particle back-transport at a membrane surface. This observation is consistent with results from other studies that indicate that the wake structure of a bubble rising in water at a low Reynolds number is larger than the size of the bubble itself [22]. For the particular bubble presented in Fig. 11, the Reynold's number was calculated to be 720 using the following equation:

$$Re = \frac{\rho d_e U}{\mu}$$

where  $\rho$  is the fluid density (980 kg/m<sup>3</sup>),  $d_e$  the bubble diameter (3.6 mm),  $U$  the bubble rise velocity (0.2 m/s) and  $\mu$  is the fluid viscosity (0.001 kg/(m s)).

When examining the shear signals in Figs. 6 and 7, the majority of the shear events were positive and similar to the one seen in Fig. 11. The results presented in Fig. 11 were for a spherical bubble rising parallel to the test fiber. However, depending on the system configuration and the gas sparging rate, other bubble geometries were also observed (Fig. 9). The geometry of the bubbles had a significant impact on the shear signals. Ellip-

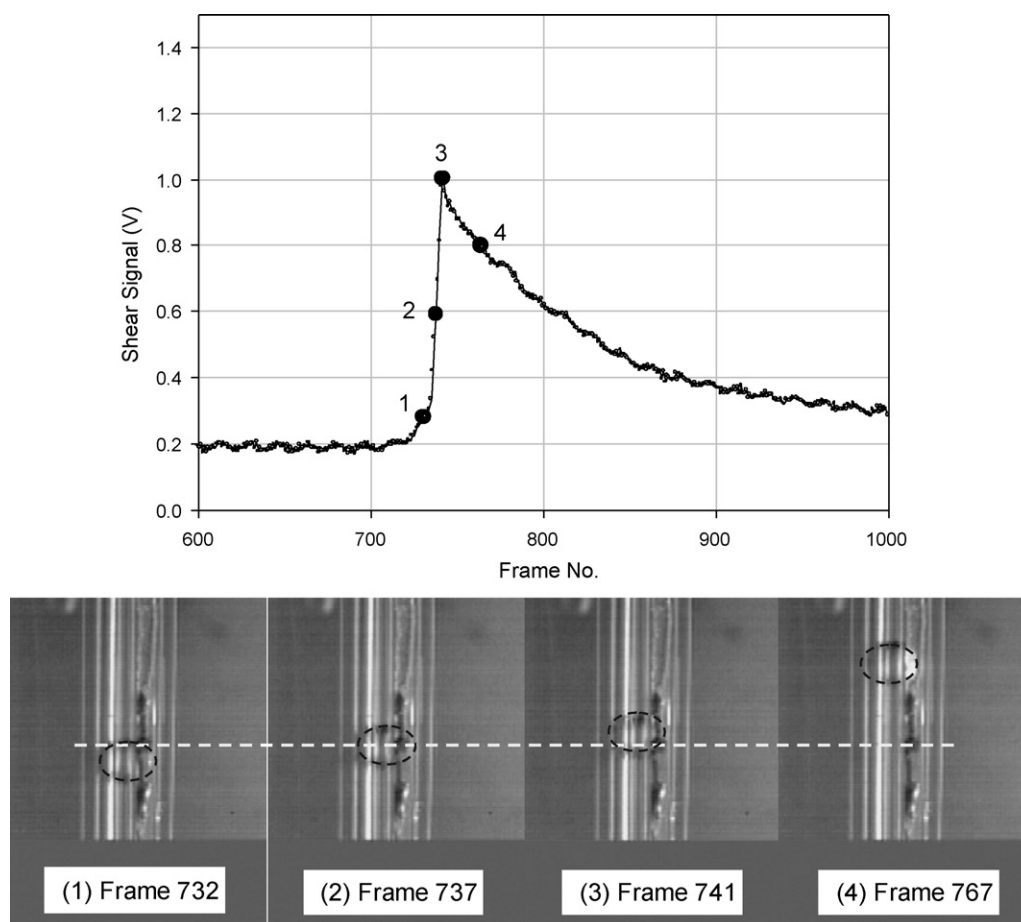


Fig. 11. Typical shear profile and images of a spherical bubble rising in tightly held fibers. The translucent vertical streaks seen in the pictures are the eight fibers in the bundle. The black dashed line highlights the bubbles rising within the fiber bundle. The white horizontal dashed line across the four images show the position of the shear probe on the test fiber (fiber packing density: medium; gas sparging rate: 0.5 mL/min; diffuser nozzle size: 2 mm).



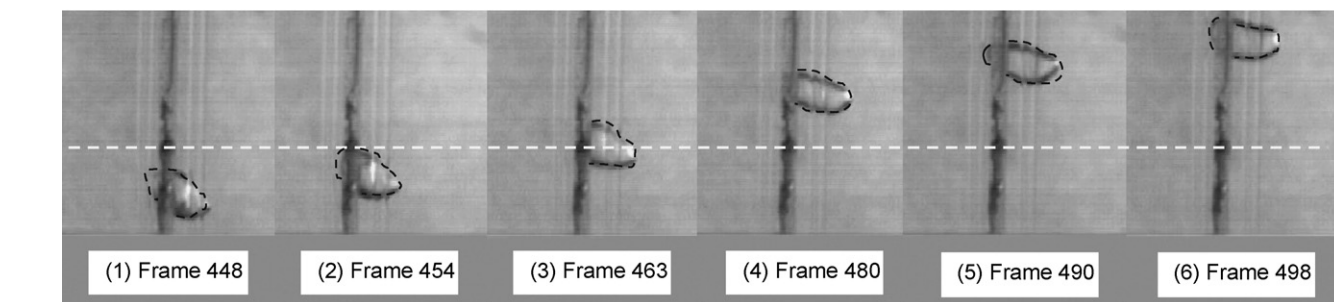
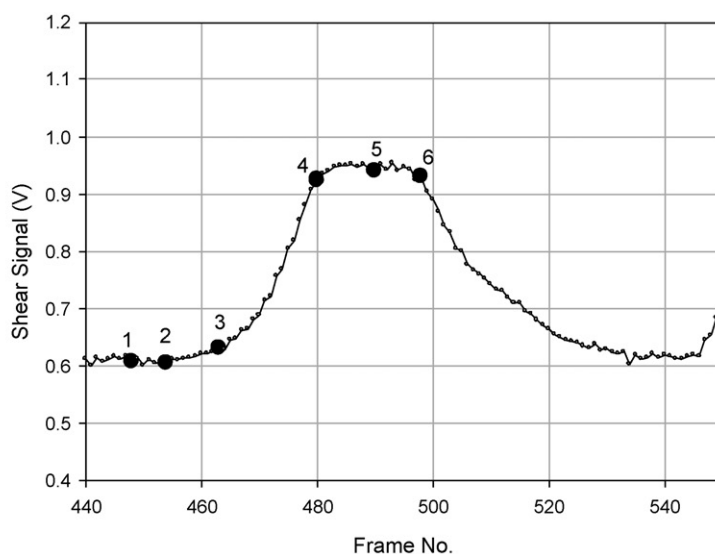


Fig. 12. Typical shear profile and images of an ellipsoidal bubble rising in tightly held fibers. The translucent vertical streaks seen in the pictures are the eight fibers in the bundle. The white horizontal dashed line across the six images show the position of the shear probe on the test fiber (fiber packing density: high; gas sparging rate: 0.5 mL/min; diffuser nozzle size: 2 mm).

soidal bubbles were commonly observed when the gas sparging rate was high. Fig. 12 shows a typical shear profile resulting from the passage of an ellipsoidal bubble by the shear probe as the bubble ‘escaped’ from a tightly held fiber bundle. Again, the maximum shear signals were observed at the tail end of the bubble (i.e. after the bubble had passed by the shear probe). However, unlike for the peak shear signal presented in Fig. 11, which lasted only 1–2 ms, the peak in the shear signal observed in Fig. 12 lasted approximately 20 ms (points 4, 5 and 6 in Fig. 12). The different shear profiles observed for the different bubble geometries likely have different effects on fouling control. Whether the high frequency of peaks similar to that seen in Fig. 11, or the longer duration of high shear signal seen in Fig. 12 are the most favorable for flux enhancement is unknown. Further research is currently underway to investigate this.

**3.1.2.2. Contribution of rising bubbles to negative peak shear forces.** As discussed earlier, one of the mechanism of fouling reduction for slug flow inside a confined membrane system has been linked to the high negative shear forces induced by the liquid flow reversal (falling film) surrounding a rising gas slug [4]. In order for flow reversal to occur, the flow path of the gas slug must be confined. As presented in Figs. 6 and 7 and Figs. A.1 and A.2 in Appendix A, negative shear signals

were periodically observed, suggesting that liquid flow reversal occurred during the passage of bubbles by the shear probe, especially for the experiments using a 2 mm diameter diffuser nozzle, a high gas sparging rate and a high fiber packing density. The number of negative shear events, compared to the total number of shear events measured for each experiment is tabulated in Table 2.

It can be seen from Table 2 that, with the exception of experiment 1-C1, the majority of the shear signals were positive in value, indicating that a reversal in the direction of liquid flow typically did not frequently occur at the surface of the test fiber. It is noted here that this trend is different than that of the shear profiles of rising gas slugs in confined systems such as tubular membranes, where each shear event consists of both positive and negative shear forces (Fig. 1). For the positive peaks, the rising bubbles were relatively close to the surface of the test fiber. However, because of the packing density and the flexibility of the fibers, as well as the different bubble sizes, the sparged bubbles were typically not completely confined to rise against the surface of the test fiber, as would be the case for a gas slug rising in a confined system. As a result, flow reversal did not consistently occur at the surface of the test fiber. However, at higher fiber packing densities and gas sparging rates, periodic flow reversal was observed. These results are consistent with the images in Fig. 9b and c, where the bubbles rising between fibers

Table 2

The number of negative peaks observed in Figs. 6 and 7

| Experiment number | Fiber packing density | Diffuser nozzle size (mm) | Gas sparging (mL/min) | Total of shear events number | Number of negative peaks |
|-------------------|-----------------------|---------------------------|-----------------------|------------------------------|--------------------------|
| 1-A1              | Low                   | 1                         | 2                     | 11                           | 1                        |
| 1-B1              | Med                   | 1                         | 2                     | 10                           | 0                        |
| 1-C1              | High                  | 1                         | 2                     | 8                            | 8                        |
| 2-A1              | Low                   | 2                         | 2                     | 10                           | 3                        |
| 2-B1              | Med                   | 2                         | 2                     | 9                            | 0                        |
| 2-C1              | High                  | 2                         | 2                     | 9                            | 0                        |
| 1-A2              | Low                   | 1                         | 10                    | 56                           | 4                        |
| 1-B2              | Med                   | 1                         | 10                    | 68                           | 1                        |
| 1-C2              | High                  | 1                         | 10                    | 24                           | 1                        |
| 2-A2              | Low                   | 2                         | 10                    | 36                           | 1                        |
| 2-B2              | Med                   | 2                         | 10                    | 49                           | 0                        |
| 2-C2              | High                  | 2                         | 10                    | 47                           | 1                        |
| 1-A3              | Low                   | 1                         | 35                    | 85                           | 0                        |
| 1-B3              | Med                   | 1                         | 35                    | 208                          | 2                        |
| 1-C3              | High                  | 1                         | 35                    | 101                          | 2                        |
| 2-A3              | Low                   | 2                         | 35                    | 43                           | 0                        |
| 2-B3              | Med                   | 2                         | 35                    | 171                          | 21                       |
| 2-C3              | High                  | 2                         | 35                    | 146                          | 45                       |

were periodically observed to be similar in shape to that of a gas slug rising in a confined tubular membrane system (Fig. 9).

As presented in Fig. 13, for the negative shear events, the shear profile was different from those presented in

Figs. 11 and 12. Unlike for the bubbles presented in Figs. 11 and 12, which were largely spherical, the bubble shown in Fig. 13 was elongated like a gas slug. The difference in bubble shapes was due to the differences in the geometry of the

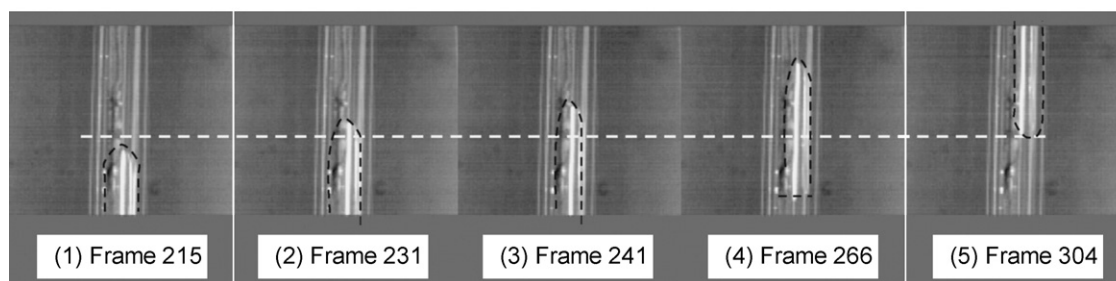
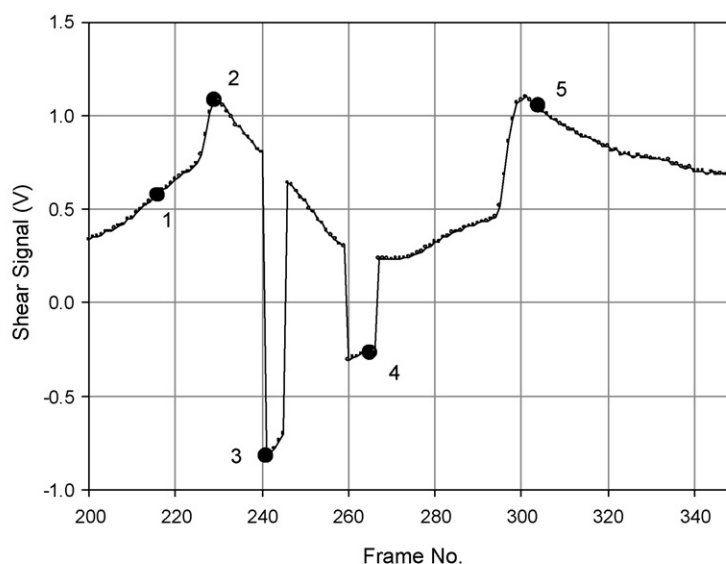


Fig. 13. Typical shear profile and images of a bubble causing negative shear peaks due to falling film surrounding the bubble. The translucent vertical streaks seen in the pictures are the eight fibers in the bundle. The black dashed line highlights the bubbles rising within the fiber bundle. The white horizontal dashed line across the five images show the position of the double probe on the test fiber (fiber packing density: high; gas sparging rate: 2 mL/min; diffuser nozzle size: 1 mm; configuration: tightly held).

flow path of the rising bubble. The bubble seen in Fig. 13 was relatively confined within a high fiber packing density bundle. Therefore, the rising bubble assumed a shape similar to that of a gas slug rising in a confined vertical tube. On the other hand, the bubbles presented in Figs. 11 and 12 were spherical because they were not confined by a high fiber packing density. Two positive and two negative peaks in shear signals were observed in Fig. 13. The first positive peak (point 2, Fig. 13) occurred as the nose of the bubble passed by the shear probe. After the first positive peak in shear signals, two negative peaks were observed at approximately 10 and 38 ms after the passage of the tip of the bubble past the shear probe. The images taken at these negative peaks in shear signals showed that the shear probe was in the falling film region surrounding the rising bubble, suggesting that the negative shear signals were due to liquid flow reversal in this region. The second positive peak in shear signals (point 5, Fig. 13) occurred near the tail end of the bubble as it rose past the shear probe. The positive shear signals that were observed between points 3 and 4 (Fig. 13) likely resulted from the non-completely confining nature of the flow path of the rising bubble and therefore a non-stable falling film. These results indicate that the negative peaks in shear signal were not due solely to liquid flow reversal in the falling liquid film region. On examination of the images taken of bubbles passing by the shear probe, negative shear signals were also periodically observed when the tail

end of rising bubbles passed by the shear probe. These results suggest that some of the negative shear signals were generated by the secondary flows induced by the wake of a rising bubble. Typical negative shear signals generated at the tail end of rising bubbles are shown in Fig. 14. As presented, positive peak shear signals (point 2 on Fig. 14) were immediately followed by a negative peak shear signal. Both the positive and negative peak shear signals occurred when the tail end of the bubble passed by the shear probe suggesting that the oscillating shear signals resulted from the secondary flows generated in the wake region of a rising bubble. It should be noted that the bubble images presented in Fig. 14 are similar to those presented in Fig. 11, but the shear profiles for the two bubbles were very different. The video imaging system used was only capable of capturing a two dimensional view of the bubble relative to the location of the shear probe. It is likely that the differences in the shear profiles presented in Figs. 11 and 14 were due to the differences in the distance between the bubbles and the shear probe in the third dimension which could not be imaged.

A closer examination of the shear profiles for the experiments in which negative shear signals were observed revealed that on average, the frequency at which negative shear signals were observed either as the nose or tail regions of a rising bubble passed by the shear probe were similar. These results indicate that negative shear events were equally likely to occur due to

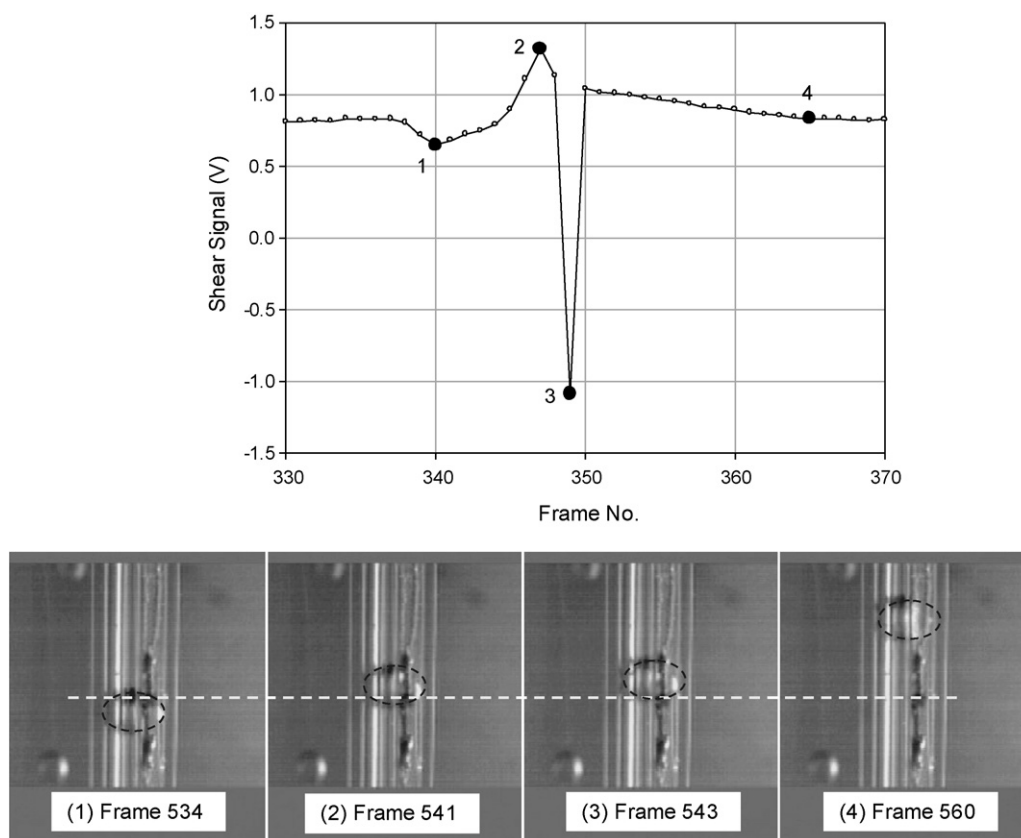


Fig. 14. Typical shear profile and images of a sparged bubble causing negative shear peaks due to flow reversal generated by instabilities in the wake region. The translucent vertical streaks seen in the pictures are the eight fibers in the bundle. The black dashed line highlights the bubbles rising within the fiber bundle. The white horizontal dashed line across the four images show the position of the shear probe on the test fiber (fiber packing density: medium; gas sparging rate: 35 mL/min; diffuser nozzle size: 2 mm; configuration: tightly held).

flow reversal or due to the secondary flows in the wake of a rising bubble.

**3.1.2.3. Shear profile surrounding the fiber.** The shear profiles shown in Figs. 6 and 7 were measured with the shear probe facing the diffuser nozzle (i.e. rising bubbles). In a separate experiment, the shear probe was rotated such that the shear probe was facing away from the rising bubbles. In general, the shear profiles obtained when rotating the shear probe away from the center of the diffuser nozzle were similar to those observed when the shear probe was facing the diffuser nozzle. However, the magnitude of the shear signals was impacted by the orientation. For both gas sparging rates investigated, the average shear signals decreased as the shear probe was rotated further away from the diffuser nozzle, as seen in Fig. 15. At the lower gas sparging rate of 5 mL/min, the average shear signal when the shear probe was rotated 180° away from the nozzle was approximately 0.18 V, while the average shear signal for the higher gas sparging rate of 30 mL/min was approximately 0.25 V. The higher average shear signal observed at the higher gas sparging rate is likely due to the higher bulk superficial liquid velocity induced by higher gas sparging rates. At a gas sparging rate of 30 mL/min, a significant decrease in shear signal (approximately three times) was observed when the shear probe was rotated 180°, relative to the diffuser nozzle. This result may provide a first step in quantifying the degree of foulant control of fibers at different locations in a full-scale submerged hollow fiber system, where the outer fibers in a bundle may experience a substantially greater amount of shear force, and therefore better fouling control, when compared to fibers located within the bundle, or the sections of a fiber that face inward and therefore away from the gas bubbles.

### 3.2. Bubble-induced shear profile for loosely held fiber bundles

Experiments with loosely held fibers at a gas sparging rate of 35 mL/min were conducted for high fiber packing density

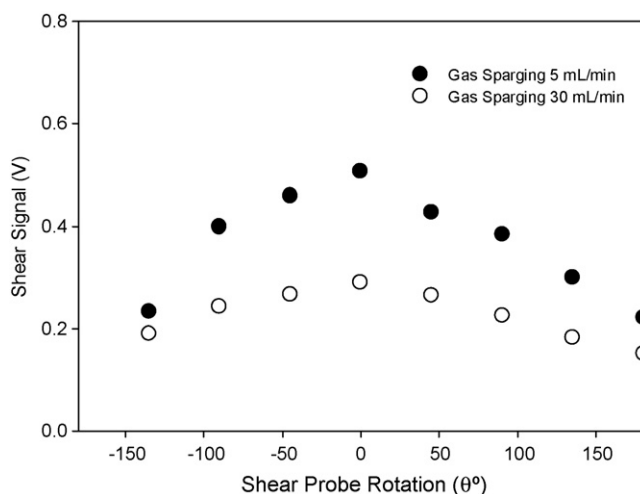


Fig. 15. Average shear signal vs. shear probe orientation at different gas sparging rates (fiber packing density: medium; diffuser nozzle size: 2 mm).

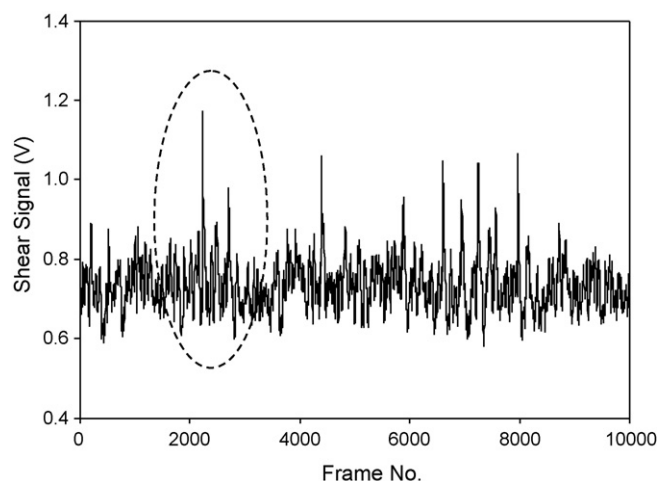


Fig. 16. Typical shear signals on loosely held fibers (fiber packing density: high; gas sparging rate: 35 mL/min; diffuser nozzle size: 2 mm). The zoomed-in view of the circled section, along with images of the rising bubble, are shown in Fig. 17.

bundles, and for diffuser nozzle sizes of 1 and 2 mm. Typical shear signals for loosely held fibers are shown in Fig. 16. All of the shear signals observed in Fig. 16 were positive indicating that for loosely held fibers, gas sparging did not result in any flow reversal. This implies that scouring of the fiber surface by a falling film, which is an important mechanism in the control of fouling in confined membrane systems [4], does not occur in loosely held submerged hollow fiber systems.

As presented in Fig. 16, the average shear signals were in general lower than those observed for the tightly held fibers (i.e. Figs. 11 and 12). However, occasionally, larger peaks in shear signal were observed. Images captured from the video recording of the experiment, shown in Fig. 17, revealed that most of the time, bubbles were not in contact with the test fibers since the flow path was not as confined as was the case for tightly held fibers. This resulted in the overall lower shear signals. Occasionally, bubbles came very close to the shear probe causing a larger peaks in shear signals. However, the frequency at which these larger peaks in shear signals occurred was relatively low. Despite this, a number of studies have indicated that higher permeate flux can typically be maintained in a loosely held submerged hollow fiber membrane system, compared to a system with tightly held fibers [23,24]. This discrepancy can likely be explained by the fact that bubbles rising in systems with tightly held fibers are relatively confined to a specific region of the fiber bundle. In contrast, for systems with loosely held fibers, the flow paths of the bubbles are not confined to a specific region.

As the bubbles rise past the loosely held fibers, secondary flows generated by the wake of the rising bubbles cause the fibers to sway. The swaying of the loosely held fibers in turn increases the likelihood that bubbles will travel to multiple regions of the loosely held fiber bundle, thus increasing the number of fibers that can benefit from the bubble-induced shear forces. Additional research using full-scale gas sparged submerged hollow fiber membranes is currently underway to further investigate this hypothesis.



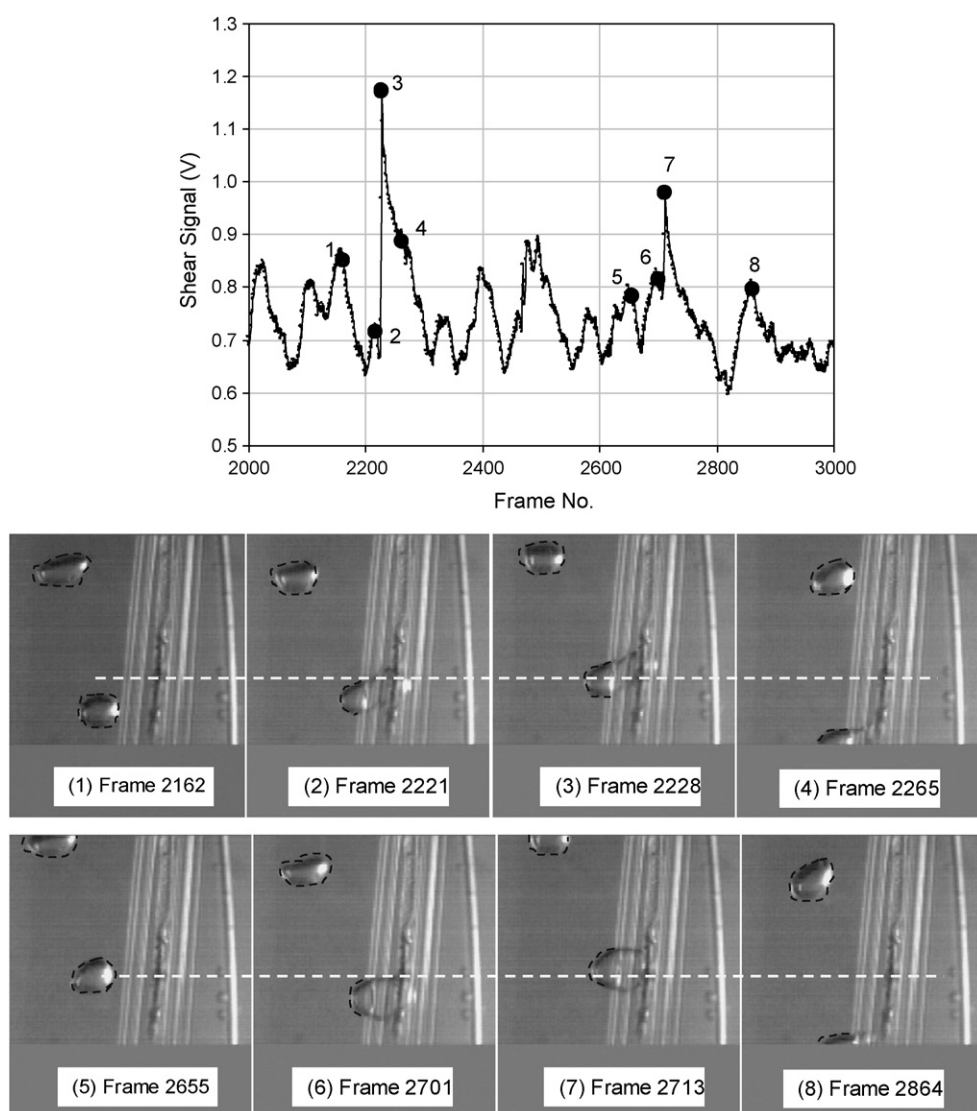


Fig. 17. The zoomed-in view of the circled section in Fig. 16, along with the synchronized images of the rising bubble. The translucent vertical streaks seen in the pictures are the eight fibers in the bundle. One of the bubbles in Frames 2221, 2228, 2701 and 2713 are partially behind the translucent fiber (fiber packing density: high; gas sparging rate: 35 mL/min; diffuser nozzle size: 2 mm; configuration: loosely held).

#### 4. Conclusions

The study results concluded that the hydrodynamic conditions in confined tubular membrane systems and unconfined submerged hollow fiber membranes are different. These results imply that, unlike confined tubular membranes, wherein one of the mechanisms by which flux enhancement is achieved is through scouring of the fiber surface by a falling film, such a mechanism does not likely significantly contribute to flux enhancement in gas sparged submerged hollow fiber membrane systems.

Shear profiles during sparging were found to be substantially affected by the operating conditions (i.e. gas sparging rates, fiber packing density, diffuser nozzle size, and module configuration). Based on the experimental data, the following conclusions were reached:

- (1) Different operating conditions yielded bubbles with different geometries (i.e. spherical, ellipsoidal or slug-like), and the different bubble geometries yielded different shear profiles.
- (2) The high number of shear events generated by small bubbles (i.e. small diffuser nozzle size) in low and medium fiber packing density bundles likely provides more favorable hydrodynamic condition for fouling reduction and flux enhancement in tightly held fibers.
- (3) Negative shear events in tightly held fibers were equally likely to occur due to flow reversal in a falling film or due to the secondary flows in the wake of a rising bubble.
- (4) Scouring of the fiber surface by a falling film, which is an important mechanism in the control of fouling in confined membrane systems, does not occur in loosely held submerged hollow fibers.

- (5) The flow path of bubbles rising in loosely held fibers were not confined to a specific region, as was the case for tightly held fibers. The larger number of fibers (i.e. regions) that can benefit from sparged bubbles in systems with loosely held fibers likely explains why higher permeate fluxes have been reported in loosely held submerged hollow fiber membrane systems compared to those in tightly held systems.

The results from the present study provide a better understanding of the hydrodynamic conditions inside submerged hollow fiber membrane modules, and therefore a better under-

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standing of possible mechanisms of fouling control via gas sparging.

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### **Appendix A**

See [Figs. A.1 and A.2](#).

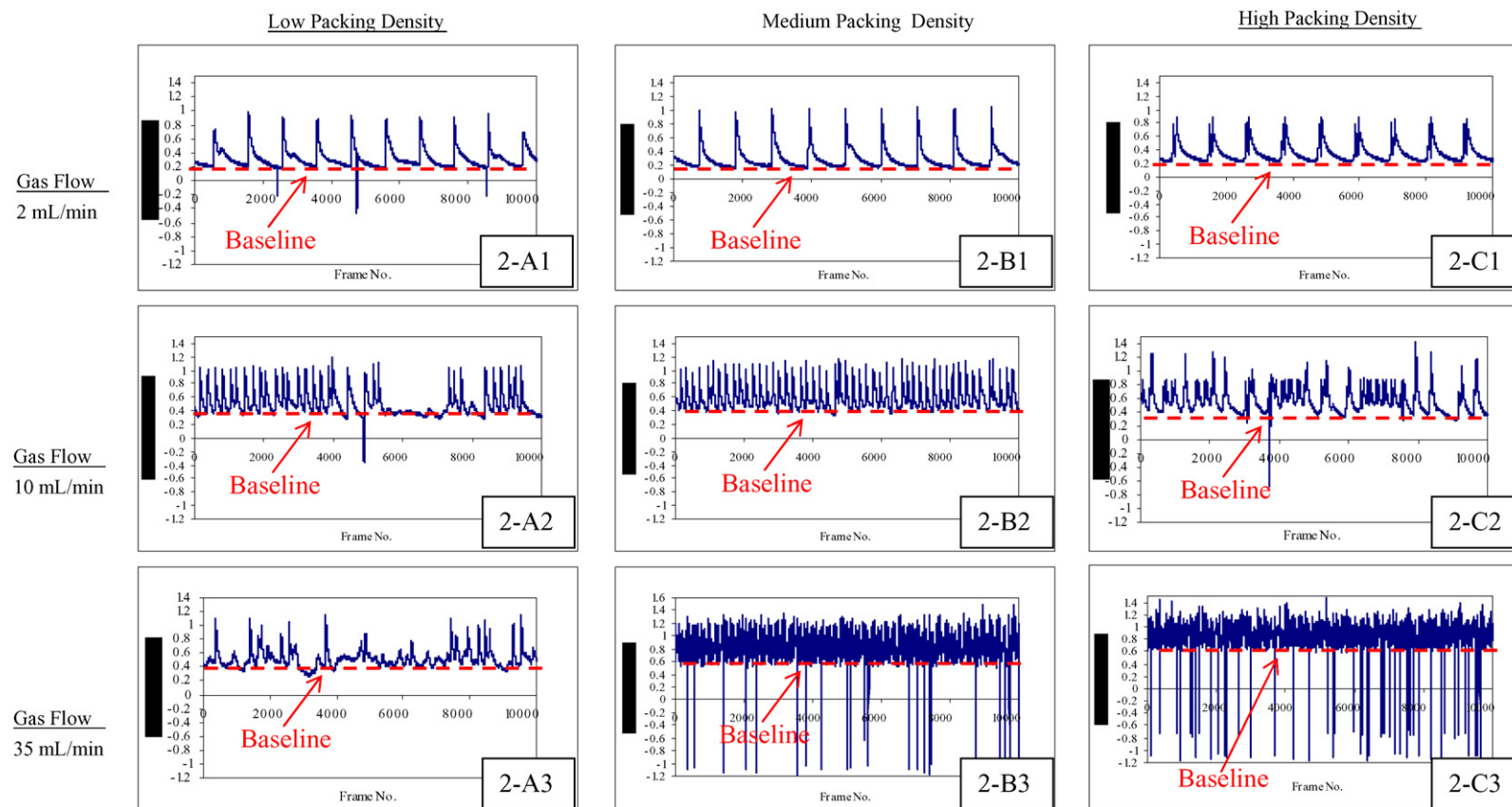


Fig. A.1. Typical shear signals on test fiber at different fiber packing densities and gas sparging rates for tightly held fibers (diffuser nozzle size: 2 mm).

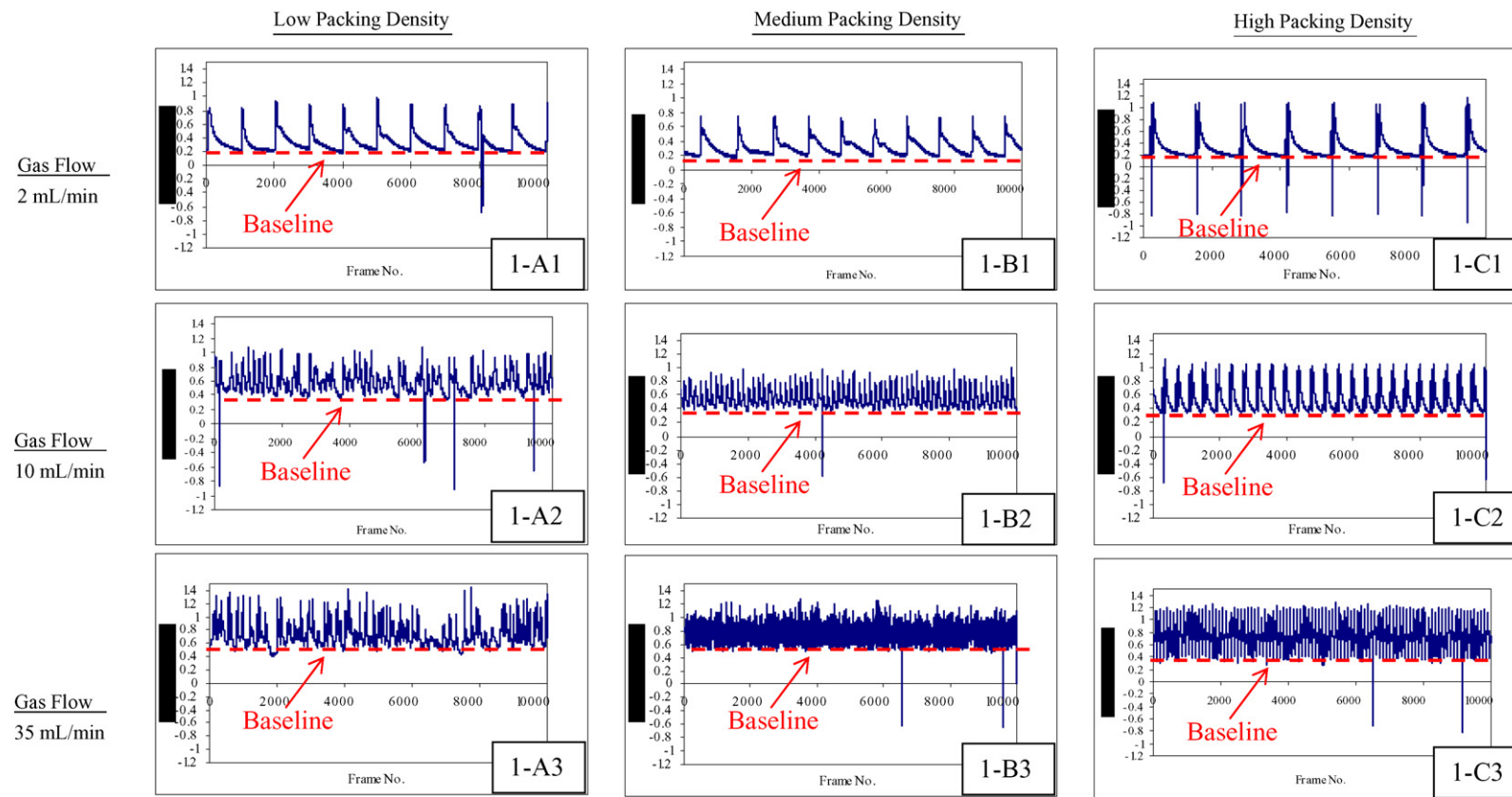


Fig. A.2. Typical shear signals on test fiber at different fiber packing densities and gas sparging rates for tightly held fibers (diffuser nozzle size: 1 mm).

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